Journal of Biomedical Informatics 164 (2025) 104792

Contents lists available at ScienceDirect

Journal of Biomedical Informatics

journal homepage: www.elsevier.com/locate/yjbin

Original Research

CECRel: A joint entity and relation extraction model for Chinese electronic medical records of coronary angiography via contrastive learning

Yetao Tong [a,c], Jijun Tong [b,c,*], Shudong Xia [d], Qingli Zhou [d], Yuqiang Shen [d]

[a] *School of Computer Science and Technology, Zhejiang Sci-Tech University, Hangzhou, 310018, Zhejiang, China*
[b] *School of Information Science and Engineering, Zhejiang Sci-Tech University, Hangzhou, 310018, Zhejiang, China*
[c] *Provincial Key Laboratory for Research and Translation of Kidney Deficiency-Stasis-Turbidity Disease, Zhejiang, China*
[d] *The Fourth Affiliated Hospital of Zhejiang University School of Medicine, Yiwu, 322000, Zhejiang, China*

ARTICLE INFO

Keywords:
Relation extraction
Data scarcity
Contrastive learning
Chinese electronic medical records
Relationship overlap

ABSTRACT

Entity and relation extraction from Chinese electronic medical records (EMRs) is a crucial foundation for constructing medical knowledge graphs and supporting downstream tasks. Chinese EMRs face challenges in accurately extracting medical entity relations due to limited data and the complexity of overlapping medical relationships. We propose CECRel, a joint extraction model for Chinese EMR entity relations based on contrastive learning and feature enhancement to address this issue. CECRel employs data augmentation strategies to generate positive and negative samples for contrastive loss computation and utilizes a feature enhancement module to enrich textual structural features, enabling the accurate extraction of complex relational triples. Experiments conducted on our constructed dataset, CACMeD, demonstrated that the model achieves an accuracy of 80.56%, a recall of 74.69%, and an F1 score of 77.51%. Furthermore, in the Baidu DuIE dataset, the model achieved an accuracy of 79.71%, a recall of 74.14%, and an F1 score of 76.82%, demonstrating that the proposed model is competitive among existing models.

1. Introduction

Structured EMRs hold significant importance in clinical medicine. However, most current EMRs are often maintained in an unstructured format, making manual information extraction challenging. This limitation prevents the full realization of EMRs' potential value, resulting in substantial resource waste [1].

Extracting information from unstructured specialized EMRs has become an urgent issue to address. Extracting key information from unstructured texts primarily relies on Natural Language Processing (NLP) technologies. NLP techniques mainly encompass Named Entity Recognition (NER) and Relation Extraction (RE). Several solutions have been proposed for extracting relational triples, such as [2,3], which utilize a pipeline approach to extract entities and relationships separately. This extraction method is prone to error propagation, as errors introduced during the entity recognition process [4] can degrade the quality of the relation extraction results. Furthermore, the useful information from the two subtasks is not fully leveraged, neglecting the implicit feature correlations between entity recognition and relation extraction. Errors that arise when identifying complex medical terminology entities can adversely affect the quality of medical relation extraction.

In light of these limitations, researchers have proposed joint entity and relation extraction. This approach can be broadly categorized into three types. The first paradigm, exemplified in [5–7], employs an end-to-end table-filling task to proxy the joint entity-relation extraction task. These methods use shared parameters to represent the correlation between entity and relation extraction; however, entities and relations are still extracted separately, leading to redundancy [8]. The second paradigm, as demonstrated in [8–11], achieves joint entity-relation extraction through sequence labeling, which requires the design of complex tagging schemes. The final paradigm, illustrated by [12–15], directly extracts triples using a sequence-to-sequence model. This approach performs well on complex relationships, such as nested entities, but is highly dependent on large-scale, high-quality training data. Data scarcity can significantly degrade the model's performance.

Due to the inherent complexity of relationships between entities in Chinese EMR texts, several instances of triple overlaps exist:

- Single Entity Overlap (SEO): A single entity may have relationships with multiple other entities.
- Entity Pair Overlap (EPO): The same pair of entities may exhibit multiple types of relationships.

Additionally, Chinese EMRs are subject to strict privacy protection and data security policies, and the standardization of EMR data is

* Corresponding author at: School of Information Science and Engineering, Zhejiang Sci-Tech University, Hangzhou, 310018, Zhejiang, China.
E-mail address: jijuntong@zstu.edu.cn (J. Tong).

https://doi.org/10.1016/j.jbi.2025.104792
Received 29 September 2024; Received in revised form 3 February 2025; Accepted 13 February 2025
Available online 24 February 2025
1532-0464/

relatively low. These factors further increase the difficulty of data acquisition and processing. This scarcity of data presents additional challenges for our research. Existing studies, such as [16,17], address the issue of data scarcity from the perspective of transfer learning. However, transfer learning methods are highly dependent on the strong correlation between the source and target domains.

Against this backdrop, our research primarily addresses "how to accurately identify entities and extract relationships from Chinese coronary angiography EMRs using a limited training set". To this end, we integrate a contrastive learning framework [18] with the Cascade Binary Tagging framework (CasRel) proposed by Wei et al. [10]. Additionally, to enhance the understanding of sentence context, we incorporate a text feature enhancement module, thereby introducing a novel method (CECRel). This approach allows for more accurate extraction of entities and relationships from EMRs. By contrasting different categories of samples, CECRel effectively improves entity and relationship extraction performance in Chinese EMRs under data-constrained conditions. Furthermore, our model demonstrates robust performance when addressing issues of relationship category imbalance and extracting overlapping triples.

This paper proposes a multi-strategy fusion approach for data augmentation that not only strengthens the dataset foundation but also diversifies sentence structures, enabling the model to learn richer feature representations. Additionally, through contrastive learning, the model acquires general data features in an unsupervised environment, thereby improving its performance with limited data. Based on the CasRel cascade encoding architecture and feature enhancement modules, this method integrates both local entity information and global sentence-level context. By iteratively contrasting positive and negative samples, the model refines its ability to distinguish relational boundaries, enhancing its performance in handling overlapping relations.

Statement of significance

Problem: Accurate Extraction of EMRs Information with Limited Data and Relational Overlap.

What is Already Known: Although there have been some studies on relation extraction from text, the results for extracting overlapping relationships and handling data scarcity have been unsatisfactory.

What this Paper Adds: This study proposes a cascading relationship extraction model that combines data augmentation and a contrastive learning framework to go a step further in data scarcity and complex relationship extraction, in addition to constructing a coronary angiography dataset.

Who would benefit from the new knowledge in this paper: Clinicians or related researchers who want to capture information accurately in limited EMRs.

2. Related work

2.1. Medical information domain

Previous research in medical information extraction has largely relied on pipeline approaches, where medical entities are identified and then linked to predefined relationship types. Early methods utilized rules, dictionaries [19], and machine learning [20,21], while advances in deep learning introduced models like LSTM and CRF. Liu et al. [22] achieved strong results in Chinese clinical entity recognition using a Bi-LSTM-CRF model, and Li et al. [23] enhanced this with a BERT-BiLSTM-CRF model tailored to Traditional Chinese Medicine (TCM) and EMRs.

LSTM models struggle with complex contextual relationships, prompting Wang et al. [24] to propose ASTRAL, an adversarial training model leveraging positional word information. For relation extraction, Qi et al. [25] developed KeMRE, combining BERT-CNN-LSTM for entity correlation and MLP for relationship prediction. However, such models require high-quality annotations and falter in low-resource scenarios. To address this, Zhang et al. [26] utilized Variational Autoencoders (VAE) for semi-supervised learning, while C. Yang et al. [27] proposed a hybrid method using residual networks, BiGRU, and attention mechanisms for extracting relationships in small EMRs.

Pipeline methods often neglect interactions between NER and RE subtasks, risking error propagation. Joint extraction models, like Sun et al.'s [28] PMEI, address this by encoding shared representations and early predictions, demonstrating effectiveness in mitigating these challenges.

2.2. General domains

NER and RE in the biomedical domain build on advancements in general domains. Cheng et al. [29] proposed a multimodal entity recognition method integrating text and images, which enhanced recognition accuracy. Yu et al. [30] introduced a span-based tagging scheme that resolves RE tasks using hierarchical boundary tagging and multi-span decoding. Zheng et al. [31] tackled overlapping challenges like SEO and EPO by proposing sequence tagging components for latent relationship prediction. Sui et al. [32] utilized a transformer-based non-autoregressive network to handle joint entity and relation extraction as a direct prediction problem. Huang et al. [33] integrated a dependency-based attention mechanism with BERT to account for word-entity dependencies, while Xu et al. [34] combined a Global Gating Mechanism (GGM) with BERT to enhance semantic representation.

Pipeline approaches optimize entity recognition and relation extraction separately, but issues like error propagation and redundancy remain. Zeng et al. [35] proposed CopyRE, a sequence generation model that decodes and copies entities to extract triples but struggles with subject-object distinction. Parameter-sharing approaches alleviate some pipeline issues but may produce redundant information due to module independence.

Joint decoding methods offer solutions to these challenges. Sun et al. [36] employed Graph Convolutional Networks (GCN) to capture entity interactions via an adjacency matrix. Sui et al. [37] proposed a Set Prediction Networks method for end-to-end joint extraction of complex data, requiring higher computational resources. Ma et al. [38] introduced a domain-specific generalized predictor that optimizes neural architecture search, achieving good performance with limited resources. Gao et al. [39] proposed a lightweight joint extraction model based on global entity matching. While effective, these joint methods face complexities that challenge implementation and scalability.

Existing methods have shown success in relation extraction but struggle with the data scarcity and the complexity of Chinese EMR. To address this, we propose CECRel, which introduces key innovations: a multi-strategy fusion data augmentation method for Chinese medical texts, combined with contrastive learning for improved feature learning in limited data. The cascade structure handles overlapping relationships, and feature enhancement better captures semantic dependencies, enabling more effective processing of Chinese medical texts.

2.3. Large language models

Large language models (LLMs) have transformed entity and relation extraction across domains, thanks to their advanced semantic understanding and knowledge integration. Notable examples include GPT [40,41], T5 [42], and LLaMA [43,44]. In biomedicine, fine-tuned LLMs excel in tasks like text mining. For example, BioBERT [45] improves biomedical text comprehension but struggles with generative tasks like summarization. BioGPT [46] demonstrates strong generative capabilities and excels in relation extraction, while TaiYi [47] fills the gap in Chinese-English biomedical NLP, achieving notable results.

However, these models perform less effectively in discriminative tasks and are prone to errors in complex scenarios involving multiple entities and relations. Furthermore, their high computational and storage demands limit their practicality for resource-constrained environments.

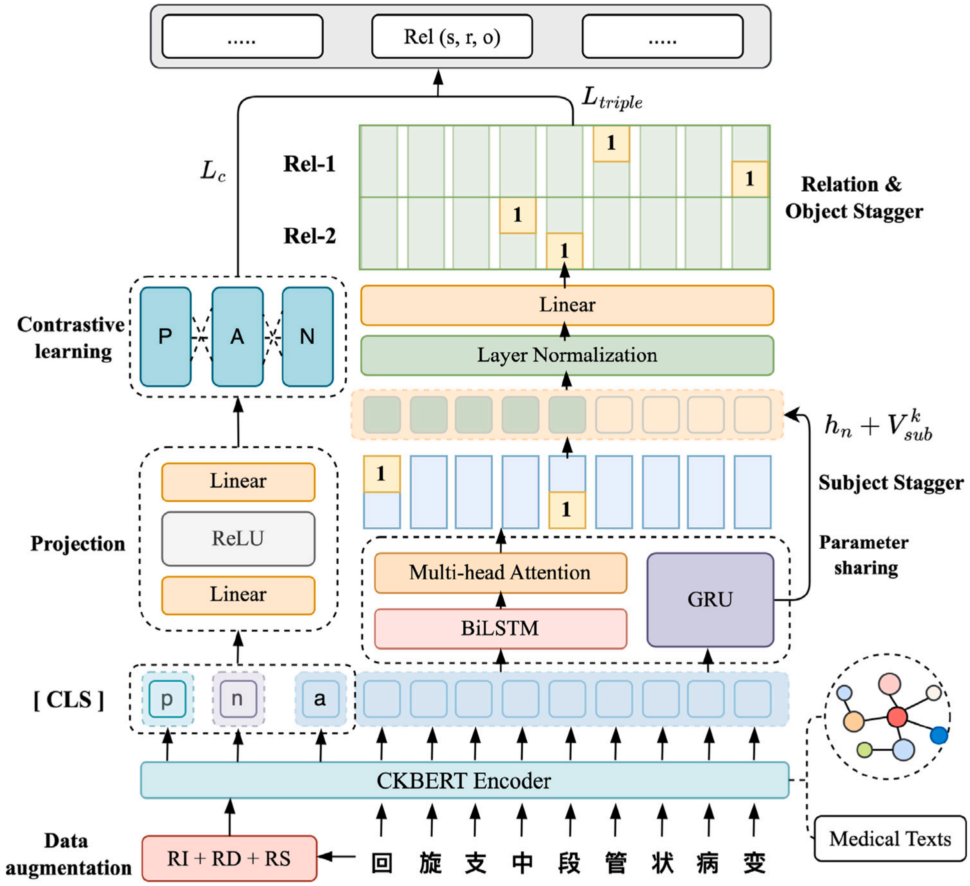

Fig. 1. Main architecture of the proposed CECRel.

3. Method

We employ the Chinese Knowledge-Enhanced BERT (CKBERT) encoder [48] to encode the text. Subsequently, the embeddings are feature-enhanced through the BMG module, followed by subject identification and object and relation prediction using CasRel [10]. Positive and negative samples are constructed through data augmentation strategies for contrastive learning. The overall architecture of the model is illustrated in Fig. 1:

During the training phase, the output of the CKBERT encoder is divided into two sub-tasks, as illustrated in Fig. 2:

- The features of positive samples ($h_p[cls]$), negative samples ($h_n[cls]$), and anchor samples ($h_a[cls]$) are projected into the same vector space using a projection head and are subsequently utilized to compute the contrastive loss.
- The CKBERT embeddings are employed in the tasks of entity and relation triple extraction.

3.1. Model structure

3.1.1. Data augmentation

By employing data augmentation strategies, positive and negative sample pairs are dynamically generated within each batch. This approach facilitates the consideration of a more diverse array of data features and enhances the generalization capability of sample pair generation. The methods for generating positive samples include: (1) **Random Deletion**: removing one-tenth of the words in the text; (2) **Random Swapping**: randomly selecting and swapping two words at the character or entity level within the text; and (3) **Random Insertion**: inserting a random Chinese word into the text.

Given that the task involves relation extraction in medical texts, the most pertinent error scenarios involve mismatches between subjects and objects as well as missing entities, which can adversely affect relation extraction. Therefore, the negative sample generation method is designed as follows: for each sample containing n pairs of relations, $x(1 \leq x \leq n)$ relation pairs are randomly selected. For each selected relation pair, the object is replaced in one of two ways with equal probability: (1) substituted with another randomly chosen entity word of a different entity type than the current object, or (2) replaced with a randomly selected non-entity word. This results in negative samples that contain logical errors while maintaining a sentence structure similar to the original text. Consequently, the model is compelled to focus on key terms and learn comprehensive and nuanced semantic information from the text.

For example, the sample “Right coronary distal segment diffuse lesions, no significant stenosis” generates the positive sample “Distal segment right coronary diffuse lesions, no significant stenosis”, preserving the “BpSy” relationship while altering word order. The negative sample, “Right coronary distal segment treated with nitroglycerin, no significant stenosis”, shifts the relationship from “BpSy” to “BpTm”. Through contrastive learning, the model learns that while syntax changes, the positive sample retains the same semantic relationship, and the negative sample represents a different relationship.

3.1.2. CKBERT encoder

The encoder module uses CKBERT, a Knowledge-Enhanced Pre-trained Language Model by Zhang et al. [48], which integrates linguistic knowledge and external information without modifying BERT’s structure [49]. By learning relationships from knowledge bases and linguistic knowledge from syntactic or dependency analyses, CKBERT improves context-aware representations. It reconstructs input sentences,

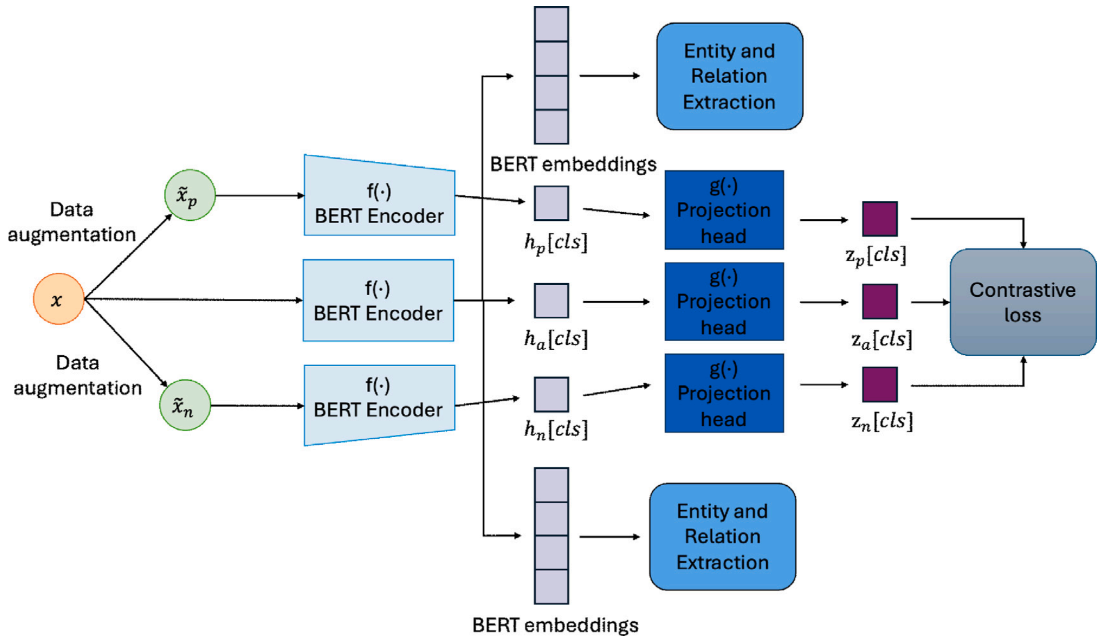

Fig. 2. Workflow of our CECRel approach.

masks additional tokens, and combines BERT's MLM task with knowledge graph augmentation. Through entity alignment and relation prediction, CKBERT enhances the understanding and reasoning of Chinese text.

We fine-tuned CKBERT for the medical domain using the EasyNLP [50] framework by injecting information from the publicly available Chinese medical knowledge graph QABasedOnMedicaKG [51] and randomly selecting 30% of the CACMeD text data. In the CKBERT model, the transformation from input embeddings to the final output involves multiple steps, processed through N stacked identical Transformer modules. Each Transformer block comprises multiple sublayers, denoted as Trans to represent CKBERT's Transformer. The processing procedure of CKBERT can be expressed as Eq. (1) and Eq. (2):

$$h_0 = SW_s + W_p \tag{1}$$

$$h_a = Trans(h_{a-1}), a \in [1, N] \tag{2}$$

where S represents the one-hot vector matrix obtained from the Chinese word segmentation of the input medical text. W_s is the word embedding matrix, and W_p is the matrix for positional information embedding. h_a represents the hidden layer vector of the ath ($a \in [1, N]$) layer, where N is the number of Transformer modules.

3.1.3. Projection head

In contrastive learning, the projection head $g(\cdot)$ is essential, particularly for self-supervised tasks using deep learning models. It maps the feature patterns of positive and negative sample pairs, $\tilde{h_p}(\cdot)$ and $\tilde{h_n}(\cdot)$, and anchor feature vectors, $\tilde{h_a}(\cdot)$, into a unified contrastive space $z(\cdot)$. This mapping improves feature representation, extracts meaningful features, and reduces computational complexity. In this study, the projection head uses a two-layer MLP: a fully connected layer (FC), a ReLU activation, and a second fully connected layer to produce $z(\cdot)$.

3.1.4. Feature enhancement

The BMG (BiLSTM-Multihead-Attention-GRU) module enhances text comprehension. Features extracted by BiLSTM capture contextual relationships and are widely used in classification tasks. The BiLSTM layer takes encoding vectors from CKBERT as input, as shown in Eq. (3). Multihead Attention captures long-range dependencies and extracts richer feature representations. Thus, a Multihead Attention layer is added to assign weights to each word, as shown in Eq. (5).

$$H = BiLSTM(h_i) \tag{3}$$

$$Attention(Q_i, K_i, V_i) = softmax\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \tag{4}$$

$$MultiheadAtt = Concat(head_1, head_2, \dots, head_h)W^O \tag{5}$$

where Q, K, and V are derived from the BiLSTM output H via three linear transformations. Each head computes the attention score by taking the dot product of Q and all K values. The outputs are concatenated, passed through a linear transformation, and produce h_n, which is input into the subject extraction module.

The GRU module is a variant of the Recurrent Neural Network (RNN). By introducing update gates and reset gates to control the flow of information [52], the model can capture contextual information within the text while adapting to relationship features in different contexts. The main equations are as follows:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \tag{6}$$

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \tag{7}$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t]) \tag{8}$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \tag{9}$$

where σ represents the sigmoid activation function, $*$ denotes element-wise multiplication, W_r, W_z, and W_h are the weight matrices corresponding to the reset gate, update gate, and candidate hidden state, respectively. h_{t-1} signifies the hidden state from the previous time step, while x_t refers to the input at the current time step.

3.1.5. Subject tagger

The Subject Tagger directly decodes the hidden layer encoding vectors h_n obtained from the BMG module. Utilizing Eq. (10) and Eq. (11), two identical binary classifiers are employed to detect the start and end positions of subjects by performing binary classification (0 or 1) for each word to determine its probability of being the start or end of a subject. A predefined threshold is set such that if the probability exceeds this threshold, the word is encoded as 1; otherwise, it is encoded as 0. Consequently, based on the positions encoded as 1 by the binary classifiers, the Subject Tagger effectively identifies all possible subjects within the input sentence.

$$p_i^{start\_sub} = \sigma(W_{start} \cdot x_i + b_{start}) \tag{10}$$

$$p_i^{end\_sub} = \sigma(W_{end} \cdot x_i + b_{end}) \tag{11}$$

where $p_i^{start\_sub}$ and $p_i^{end\_sub}$ represent the probabilities of the ith token in the input vector being the start and end of the subject, respectively. W_{start} represents the training weights, b_{start} is the bias term, and σ is the sigmoid activation function. Eq. (12) is used to determine the span and position of the subject in x.

$$p_\theta(s \mid x) = \prod_{t\in\{start\_sub,end\_sub\}} \prod_{i=1}^{L} (p_i^t)^{I=\{y_i^t=1\}}(1-p_i^t)^{I\{y_i^t=0\}} \tag{12}$$

where L represents the length of the text. $y_i^{start\_sub}$ denotes the binary label for the start position of the subject for the ith token in x, and similarly, $y_i^{end\_sub}$ represents the label for the end position of the subject. If the position is correct, the label is set to 1; otherwise, it is set to 0. The parameters are denoted as $\theta = \{W_{start}, b_{start}, W_{end}, b_{end}\}$.

3.1.6. Relation-specific object tagger

After layer normalization, the model proceeds with object relation tagging. The high-level tagging module simultaneously identifies objects and the relationships corresponding to the subjects and objects, as illustrated in Eq. (13) and Eq. (14). Although it shares the same structure as the subject tagging layer, the relation object tagging layer differs in that, unlike the subject tagging layer which directly decodes the encoding vector h_n, it incorporates the features of the subject.

$$p_i^{start\_obj} = \sigma(W_{start}^r(x_i + v_{sub}^k) + b_{start}^r) \tag{13}$$

$$p_i^{end\_obj} = \sigma(W_{end}^r(x_i + v_{sub}^k) + b_{end}^r) \tag{14}$$

where $p_i^{start\_obj}$ and $p_i^{end\_obj}$ represent the probabilities of identifying the ith word in the input vector as the start and end positions of the object, respectively. v_{sub}^k represents the vector representation of the encoding detected as the kth subject object in the subject labeling module. Eq. (15) derives the probability of extracting the object from the sentence vector that integrates the subject features for a specific relationship:

$$p_{\phi r}(o \mid s, x) = \prod_{t\in\{start\_obj,end\_obj\}} \prod_{i=1}^{L} (p_i^t)^{y_i^t}(1-p_i^t)^{1-y_i^t} \tag{15}$$

where L represents the text length, $y_i^{start\_obj}$ is the binary label for the start position of the subject in the ith token of x, and $y_i^{end\_obj}$ is the label for the end position. If the position is correct, the label is set to 1; otherwise, it is set to 0. The parameters are $\phi_r = \{W_{start}^r, b_{start}^r, W_{end}^r, b_{end}^r\}$.

3.2. Loss function

3.2.1. Contrastive loss

This aims to maximize the similarity between positive sample pairs within the same vector space while simultaneously minimizing the similarity between the anchor and a batch of negative samples in the same vector space.

$$L_c = -log\frac{exp(sim(z_i, z_i^+)/\tau)}{\sum_{k=1}^{k} exp(sim(z_i, z_k)/\tau)} \tag{16}$$

where z_i and z_i^+ are the projected representations of the positive sample pairs, while $sim(u, v)$ denotes the similarity between u and v. τ is the temperature coefficient, adjusting the similarity scale. z_k represents the projected representation of any sample in the pool, positive or negative.

3.2.2. Triple extraction loss

Given a vector representation x_i of a sample from the training set D, we extract a series of relation triplets $T_i = (subject, relation, object)$. The objective is to maximize the likelihood of the training dataset's data:

$$\begin{aligned} &\prod_{i=1}^{|D|}\left[\prod_{(s,r,o)\in T_i} p((s,r,o) \mid x_i)\right] \\ &= \prod_{i=1}^{|D|}\left[\prod_{s\in T_i} p(s \mid x_i) \prod_{(r,o)\in T_i} p((r,o) \mid s, x_i)\right] \\ &= \prod_{i=1}^{|D|}\left[\prod_{s\in T_i} p(s \mid x_i) \prod_{r\in T_i|s} p_r(o \mid s, x_i) \prod_{r\in R\backslash T_i|s} p_r(o_\phi \mid s, x_i)\right] \end{aligned} \tag{17}$$

where $|D|$ represents the training set size, $s \in T_i$ denotes the subject in triplet T_i, $(r, o) \in T_i$ is the object-relation pair. T_i, $r \in T_i \mid s$ represents the relationship in triplets where s is the subject, and $r \in R\backslash T_i \mid s$ represents relationships not involving s. o_ϕ is the "null" object, indicating that the subject lacks a corresponding trailing entity. The loss value is the sum of the losses from the subject and object extraction tasks under specific relationship conditions, as shown in Eq. (18).

$$L_{triple} = \sum_{i=1}^{|D|}\left[\sum_{s\in T_i} logp_\theta^{sub}(s \mid x_i) + \sum_{r\in T_r|s} logp_\theta^{obj}(o \mid s, x_i)\right] \tag{18}$$

where T_r represents all relationships with s as the subject.

The final loss is obtained by adding the weighted sum of the contrastive loss and the triple extraction loss, as shown in Eq. (19):

$$L = W_{re} \cdot L_{triple} + W_{span} \cdot L_c \tag{19}$$

where W_{re} represents the weight for the triple extraction loss, W_c represents the weight for the contrastive loss.

4. Experiments

4.1. Dataset

4.1.1. Private dataset

The CACMeD private dataset comprises 571 coronary angiography surgical records obtained from a top-tier hospital in Zhejiang, China. Each surgical record documents the detailed diagnostic and treatment processes undertaken by physicians during coronary angiography procedures, including patient medical history, surgical details, and postoperative evaluation results. Our data were approved by the hospital's ethics committee, and all patient information was anonymized prior to use to ensure privacy and data security. To enhance the model's performance, we performed text preprocessing on the original surgical records, which included removing meaningless symbols and correcting typographical errors. Following data cleaning, we employed the BIOES (Begin, Inside, Outside, End, Single) sequence labeling method for annotation [53]. This process resulted in the creation of a Chinese coronary angiography surgical records dataset.

The final dataset consists of 4,671 samples, each containing one or more entity pairs and their corresponding relationships. The maximum sequence length is 243 tokens, with an average sequence length of 40 tokens. We defined six entity types and eight relationship types as follows:

Entity Types and Counts:

Body Parts (Bp): 5,037 entities (27.59%); Symptoms and Signs (Sy): 6,226 entities (34.11%); Medical Examination (In): 703 entities (3.85%); Treatment Mode (Tm): 3,161 entities (17.31%); Medication and Dosage (Do): 1,203 entities (6.59%); Medical Consumables (Mc): 1,922 entities (10.53%).

Relationship Types and Counts:

Symptoms of Body Parts (BpSy): 5,379 relationships (44.48%); Examination of Body Parts (BpIn): 72 relationships (0.59%); The Way Body Parts Are Treated (BpTm): 2,671 relationships (22.09%); The

Drug and Dosage Used in the Treatment (TmDo): 1,199 relationships (9.91%); Medical Consumables for Treatment (TmMc): 970 relationships (8.02%); Symptoms Found by Medical Examination (InSy): 829 relationships (6.86%); Medical Consumables for Medical Examinations (InMc): 528 relationships (4.37%); Medical Consumables for Body Parts (McBp): 444 relationships (3.67%).

Additionally, we analyzed the occurrences of SEO and EPO within the dataset, finding 4,064 SEO cases and 88 EPO cases, alongside 7,940 normal relationship instances. Despite encompassing multiple relationship types, the dataset exhibits class imbalance, with the BpSy category being significantly more prevalent than other relationship categories.

4.1.2. Public dataset

We conducted experiments on the publicly available DuIE dataset released by Baidu [54]. This large-scale, high-quality Chinese information extraction dataset contains 211,529 samples, defines 49 relationship types, and includes 82,413 SEO cases, 22,174 EPO cases, and 106,942 cases of other relationships. Evaluating our model on this dataset further confirms its performance on diverse and extensive data.

For fine-tuning the CKBERT model, we used the open-source QABasedOnMedicalKG dataset by Liu Huangyong [51]. This disease-centric biomedical knowledge graph includes 44,111 entities, 294,149 relationship pairs, seven entity types, ten relationship types, and eight attributes.

4.2. Evaluation metrics

Triple extraction necessitates the accurate identification of subjects, objects, and relationship types. To evaluate the performance of our model, we employ three key performance indicators: Precision, Recall, and F1-Score.

$$Precision = \frac{\sum_c TP_c}{\sum_c TP_c + \sum_c FP_c} \tag{20}$$

$$Recall = \frac{\sum_c TP_c}{\sum_c TP_c + \sum_c FN_c} \tag{21}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{22}$$

where c represents the relation label, TP denotes the number of correctly extracted triplets, FP refers to the number of incorrectly extracted triplets, and FN indicates the number of cases where the triplet exists but was not correctly identified by the model. A triplet relation (s,r,o) is considered correct only when the subject (s), object (o), and relation (r) are all accurately identified.

4.3. Implementation details

The experiments were conducted on a server configured with an AMD 5900X CPU, an NVIDIA GeForce RTX 3090 GPU, and 64 GB of memory. The deep learning environment was established using Python version 3.9 and PyTorch version 1.13.1. In an NVIDIA RTX 3090 GPU environment, CECRel, due to the addition of contrastive learning logic, increases training time by 10%–20% compared to CasRel, while inference time remains nearly the same. On lower-end GPUs, such as the 1080Ti, CECRel still runs stably. To minimize bias and fully utilize the available data, we employed K-fold cross-validation in this study. Additionally, to balance training time with comprehensive data utilization, we set the value of K to 5, thereby better maintaining the stability of the experimental results.

For the CECRel model, we set the learning rate to 1×10^{-5}, the maximum sequence length to 300, the number of epochs to 50, and the text encoder's hidden dimension to 768. We employed the Adam optimizer with a batch size of 8. Additionally, we configured the temperature parameter τ to 0.1, and both the re-weighting factor and span-weight to 0.5.

In our experiments, learning rates above 2×10^{-5} caused instability, while rates below 5×10^{-6} led to slow convergence. A rate of 1×10^{-5} balanced both. For the temperature τ, values below 0.07 resulted in overly distinct samples, while values above 0.2 caused insufficient feature differentiation. A temperature of 0.1 achieved the best balance. We explored re_weight and span_weight values in the range $[0.3, 0.7]$, with 0.5 providing the optimal balance. A batch size of 8 ensured stable training while respecting GPU memory limits.

The implementation details for the baseline model CasRel [10] are as follows: a learning rate of 1×10^{-5}, maximum sequence length of 300, 20 epochs, text encoder hidden dimension of 768, Adam optimizer, and a batch size of 8.

Furthermore, for the CKBERT fine-tuning tasks, our experimental parameter settings are detailed below: a learning rate of 5×10^{-5}, 5 epochs, a sequence length of 256, and a batch size of 16.

5. Results and analysis

5.1. Relation extraction

In order to verify the validity of the CECRel model, experiments were conducted on the CACMeD and DuIE datasets. A comparative analysis was performed between CECRel and the following models:

SpERT [55]: A span-based model that incorporates negative sample features during embedding for joint entity and relation extraction. OneRel [56]: A model combining a score-based classifier and relation-specific tagging for joint extraction. BiRTE [57]: A token-based model for bi-directional entity and relation extraction using shared encoders. TPLinker [11]: A model with a handshake tagging scheme for multi-type boundary tagging. EmRel [58]: A model that captures triple associations by modeling entities, contexts, and relationships, using representation permutations for extraction. AGT [59]: A span-based transformer encoder–decoder with a pointing mechanism. CasRel [10]: A binary tagging framework for entity extraction, relation identification, and triple extraction. Table 1 presents the experimental results of each model in CACMeD and DuIE.

Table 1 shows that the proposed CECRel model outperforms the baseline CasRel on the CACMeD dataset, improving the F1 score by 6.44%, precision by 7.54%, and recall by 5.46%. It also demonstrates competitive results against other models, confirming its superior triple extraction capabilities.

The enhancements result from data augmentation and contrastive learning. Aligning similar samples while separating distinct ones improves feature discrimination. Augmenting minority class samples balances positive and negative pairs, addressing class imbalance and boosting robustness in extracting relationships for underrepresented classes.

For the DuIE dataset, the proposed CECRel model demonstrates competitive performance. Specifically, the Precision metric is 3.32% lower than that of the BiRTE model, while the Recall metric is 1.15% lower than that of the EmRel model. Although the F1-Score remains relatively competitive, the overall improvement is less significant for this dataset. This discrepancy occurs because CECRel's data augmentation strategy is specifically designed for medical texts. The syntactic structures and predominantly short texts in the DuIE dataset differ significantly from those in medical texts, which limits the effectiveness of the BMG module and, as a result, reduces performance gains.

Additionally, due to the similar data characteristics of the DuIE and CACMeD datasets, even though they belong to different domains, the DuIE dataset also contains multiple entity pairs and diverse relationships per sample, with numerous overlapping relations such as SEO and EPO. Both datasets also have a certain proportion of noisy annotations. As a result, the proposed model may be similarly affected, which explains why its performance on the DuIE dataset does not show a significant decline.

To validate the ability of the proposed CECRel model to extract overlapping triples, we conducted extensive experiments on different types of sentences. We categorized the sentence types into SEO,

Table 1
Comparative experiments.

Model	DuIE			CACMeD(Ours)		
	F1	P	R	F1	P	R
SpERT	71.61	80.45	64.52	65.48	62.84	68.35
OneRel	74.97	76.28	73.71	66.30	80.37	56.42
BiRTE	74.34	**83.03**	67.29	70.48	71.35	69.64
TPLinker	73.80	75.31	72.35	72.53	71.82	73.26
EmRel	76.33	77.40	**75.29**	73.51	72.63	74.41
AGT	76.12	78.34	74.02	75.35	77.18	73.62
CasRel	73.04	74.32	71.81	71.07	73.02	69.23
CECRel(Ours)	**76.82**	79.71	74.14	**77.51**	**80.56**	**74.69**

Table 2
Results of extraction performance with or without overlapping triples.

Model	KPIs	Overlap scenarios		
		SEO	EPO	Normal
CECRel	F1	**72.81**	**71.62**	81.43
	Precision	75.13	74.18	82.35
	Recall	70.62	69.24	80.53
CasRel	F1	69.73	70.55	75.41
	Precision	71.74	69.72	77.33
	Recall	67.91	71.39	73.68
ATG	F1	68.63	66.43	**82.49**
	Precision	70.93	67.43	84.57
	Recall	66.48	65.47	80.51
EmRel	F1	67.57	65.45	78.94
	Precision	67.28	66.31	79.35
	Recall	67.88	64.63	78.55
OneRel	F1	70.70	68.77	76.20
	Precision	71.68	69.46	79.31
	Recall	69.75	68.11	73.34
Spert	F1	57.95	61.06	72.86
	Precision	58.35	61.46	73.52
	Recall	57.57	60.68	72.22

EPO, and Normal. Table 2 presents the experimental results of CECRel compared to the baseline model CasRel on the CACMeD dataset. The experiments indicate that CECRel shows notable improvements in handling overlapping triples.

This improvement is attributed to the introduction of a cascade decoder architecture in our model, which divides the entity and relation extraction process into two subtasks. First, the model predicts the subject entity and its corresponding relation. Then, starting from the subject entity, it independently predicts each object entity for that subject, rather than predicting all relations simultaneously across the entire sentence. This approach allows the same subject to be reused across multiple relations without mutual interference. For example, in the sentence "右冠远段弥漫性病变，无明显狭窄。" ("Diffuse lesions in the distal right coronary artery, no significant stenosis".), the model performs subject decoding twice, resulting in the triples:

- <Distal Right Coronary Artery, BpSy, Diffuse Lesions >
- <Distal Right Coronary Artery, BpSy, No Significant Stenosis >

This mechanism effectively prevents interference between different entity pairs and their corresponding relations.

Furthermore, by employing data augmentation techniques, we constructed several negative samples that contained logical errors or textual errors while maintaining structural types. This strategy forces the model to focus more on keyword features, thereby enhancing its ability to distinguish complex relationships.

In the aforementioned two experiments, the proposed CECRel model consistently demonstrated highly competitive results. While the CasRel model also employs a cascade decoder approach, it does not effectively manage the subtle differences inherent in overlapping relations. In contrast, our model leverages a sophisticated design of positive and negative samples and employs contrastive loss, enabling the model to better learn fine-grained semantic distinctions and significantly enhancing its capacity to handle complex relational details.

CECRel outperforms the ATG model, with improvements of 4.18% and 6.17% in F1-Score for SEO and EPO scenarios, respectively. In overlapping relation tasks, capturing global interactions among entities and relations is crucial, but the autoregressive nature of the ATG model limits this ability. CECRel addresses this by using the attention mechanism in the BMG module to model long-range dependencies and focus on key global information. Parameter sharing at the encoder level and feature interactions at different stages ensure global consistency and a comprehensive understanding of features.

The EmRel model predicts triples by aligning entity and relation representations through Tucker [60] decomposition. While Tucker decomposition models triple relationships, it loses fine-grained features, reducing its ability to distinguish overlapping relations. Our proposed model addresses this by using repeated contrasts between positive and negative samples, helping the model better learn the discriminative boundaries of relationships.

For the OneRel model, entity pairs are marked separately within different relation submatrices, and interactions between entities are modeled based on the HOLE mechanism [61]. This can lead to inconsistent markings across matrices, potentially resulting in the loss of fine-grained information and negatively impacting the recognition of complex relationships.

The proposed model's cascade decoding mechanism independently executes different subtasks, allowing the same entity to naturally associate with multiple relations without mutual interference. Furthermore, our method extends beyond relation-level contrast to support entity-level contrast, thereby further enhancing the model's ability to distinguish different types of relations within the same entity pair.

5.2. Entity recognition

To assess the CECRel model's effectiveness in recognizing medical entities, we analyzed various categories (Table 3). Accurate identification requires both start and end positions to be correct. The "In" and "Mc" categories exhibit lower F1 scores due to smaller sample sizes.

Table 3
Results of model recognition performance for different entities.

Model	KPIs	Entity					
		Bp	Sy	In	Tm	Do	Mc
CECRel	F1	82.62	82.57	79.81	84.01	89.89	80.81
	Precision	86.19	83.74	86.59	89.13	95.85	85.91
	Recall	79.33	81.43	74.02	79.44	84.62	76.28
CasRel	F1	82.05	82.01	75.71	80.15	88.44	79.18
	Precision	85.44	84.20	80.42	85.98	94.42	82.11

The entity definitions in the table refer to those listed in Section 4.1.1.

Table 4
Results of different data augmentation strategies in different data volumes.

Volume	Strategy	Relation(F1)							
		BpSy	BpIn	BpTm	TmDo	TmMc	InSy	InMc	McBp
100%	–	96.26	51.59	94.76	73.14	71.37	69.03	65.54	63.19
	RD	95.14	52.04	94.77	72.43	69.95	69.81	63.08	62.06
	RS	96.84	54.54	95.80	74.18	72.76	70.26	66.15	64.81
	RI	96.49	53.64	94.26	73.61	71.03	69.01	65.91	63.98
	RD+RS	98.62	59.18	95.92	75.72	75.74	71.03	68.58	67.85
	RD+RI	96.74	55.22	94.52	74.11	69.81	68.67	65.43	63.02
	RS+RI	97.63	58.72	95.72	76.87	74.19	72.53	69.77	68.89
	RD+RS+RI	97.96	61.25	96.02	76.31	74.34	73.93	70.84	69.43
65%	–	90.34	44.93	85.03	65.87	62.01	61.42	58.52	56.20
	RD	91.38	44.03	86.48	66.33	62.94	61.95	59.82	56.91
	RS	93.14	47.77	88.47	68.31	65.19	63.90	60.40	57.94
	RI	92.47	46.93	88.16	67.85	64.24	62.36	60.13	58.02
	RD+RS	95.36	48.36	93.84	72.27	69.53	68.30	64.14	61.56
	RD+RI	93.67	46.35	89.78	68.13	65.44	63.09	61.39	58.63
	RS+RI	95.69	48.78	94.02	72.87	70.13	67.63	64.19	61.25
	RD+RS+RI	95.76	52.99	94.26	73.53	71.21	68.37	66.42	64.82
50%	–	88.74	39.35	82.79	62.98	59.81	59.13	56.77	53.87
	RD	89.37	40.16	83.02	61.92	60.29	60.01	56.90	53.29
	RS	91.03	43.50	84.78	63.01	62.47	62.69	58.22	56.22
	RI	90.45	42.80	83.48	62.80	61.83	61.35	57.41	55.32
	RD+RS	93.42	47.53	87.83	67.59	64.72	63.31	60.25	57.91
	RD+RI	90.57	42.13	83.09	63.88	61.07	59.43	55.93	53.98
	RS+RI	92.04	44.54	86.84	66.11	62.84	61.86	59.79	56.14
	RD+RS+RI	93.59	49.15	90.15	70.29	68.33	66.93	65.93	62.27

RD stands for random deletion, RI stands for random insertion, and RS stands for random swap.

CECRel shows notable improvement in recognizing "In" entities over CasRel, demonstrating the power of contrastive learning in low-resource settings. By generating positive and negative sample pairs, the model learns discriminative features without relying heavily on large datasets, reducing the need for manual annotations and enhancing overall performance.

Additionally, the feature enhancement module enables the model to handle noise, long texts, and complex structures effectively, ensuring robust performance on intricate medical records.

5.3. Impact of data augmentation methods on triple extraction

To evaluate the effects of various data augmentation techniques on the model's ability to extract triples, the influence of these methods on different types of relationships, and their performance across datasets of varying sizes, we implemented seven distinct data augmentation strategies. These strategies were applied to datasets scaled to 100%, 65%, and 50% of their original sizes, respectively. The dataset partitioning strategy involved randomly subsampling the data to achieve the target sizes. The experimental results are summarized in Table 4.

The experimental results demonstrate that, compared to the baseline without data augmentation, all three data augmentation methods enhanced triple extraction performance. Notably, the random swapping strategy achieved more significant improvements than the random deletion method. Furthermore, combinations of the three data augmentation strategies generally outperformed single strategies or the absence of data augmentation.

Our approach outperforms traditional augmentation methods like synonym replacement or back-translation in preserving complex relationships between medical entities. While conventional techniques may disrupt overlapping relationships, our strategy introduces useful changes while maintaining entity pair integrity.

Due to the imbalanced distribution of relationship types in the dataset, the number of instances for the "BpTm" relationship type is significantly smaller than for other relationship types. Since the gradient updates of the loss function depend on sample frequency, the gradients for minority class samples are diluted, causing the model to form decision boundaries biased toward majority classes and neglect features of minority classes. Consequently, the performance improvements for these types are not as prominent. In contrast, the relationship types "BpSy" and "BpTm" have larger proportions, resulting in significantly superior relationship extraction performance in the experiments.

The effects of data augmentation on different relationship types stem from several factors. For majority classes like "BpSy", improvements arise from contrastive learning that captures fine distinctions between similar patterns. For minority classes like "BpIn", random swapping and insertion help diversify contexts, addressing sample scarcity and balancing the class distribution. This is particularly important for overlapping relationships, where maintaining semantic consistency is key.

In experiments with the full-size dataset, the "BpSy" relationship type showed an improvement of 1.7% in F1 score using the RD+RI+RS data augmentation strategy compared to the baseline. Meanwhile, the rare relationship types "BpIn", "McBp", and "InMc" improved by 9.66%, 6.24%, and 5.30%, respectively, demonstrating more pronounced enhancements. Similarly, in experiments with datasets sized at 65% and 50%, the same pattern was observed. This indicates that our proposed data augmentation methods enable more comprehensive learning of features for rare relationship types across different contexts.

Table 5
Ablation experiments.

Methodologies	F1	P	R
CECRel(Ours)	77.51	80.56	74.69
-CL	74.43	76.81	72.30
-BiLSTM	77.05	79.82	74.17
-Multi-Attn	76.83	78.89	73.91
-GRU	76.92	79.03	74.02
-BMG	75.91	78.77	73.25
-CL -BMG	71.07	73.02	69.23

Specifically, under the influence of contrastive learning, the feature space between rare and majority classes is more distinctly separated, enhancing the model's discriminative capability and thereby addressing data scarcity and class imbalance to some extent.

Additionally, in experiments with the 65% sized dataset, the F1 score using the best data augmentation strategy increased by 0.31% compared to the baseline on the full-size dataset. In the 50% sized dataset experiments, the F1 score using the best data augmentation strategy increased by 5.29% compared to the baseline in the 65% sized dataset. These two sets of experiments demonstrate that our data augmentation strategies can effectively enhance the model's relationship extraction performance in small-sample scenarios.

5.4. Ablation study

An ablation study was conducted to further ascertain the effects of the model components. The experimental results are shown in Table 5.

From the results in the table, it is evident that removing the contrastive learning framework from our CECRel model leads to a decrease of 3.08% in the macro-average F1 score on the CACMeD dataset. This decline underscores the critical role of contrastive learning in enhancing triple relation extraction and improving performance on minority class extractions. Additionally, eliminating the BMG feature enhancement module results in a 1.6% reduction in the F1 score on the CACMeD dataset. Specifically, removing BiLSTM led to a 0.46% F1 drop, indicating its role in sequence modeling. Removing Multi-Attention caused a 0.68% decrease, underscoring its importance for long-range dependencies. Removing GRU resulted in a 0.58% decline, showing its contribution to capturing semantic associations in long sequences.

5.5. Comparison with large language models

While large language models have achieved remarkable results in general relation extraction tasks, our experiments demonstrate that specialized architectures like CECRel can deliver competitive performance in the medical domain, particularly when dealing with limited data and overlapping relations. Compared to large models, our approach offers several key advantages: Domain Adaptability: Through targeted feature enhancement and contrastive learning, the model better adapts to the specific characteristics of medical text. Resource Efficiency: The architecture requires significantly fewer parameters and computational resources than large models, making it more practical for clinical applications. Data Efficiency: Our method achieves robust performance with limited training data, whereas large models typically require vast amounts of data to reach optimal performance.

5.6. Clinical applications

The CECRel model, with its strong performance in entity-relation extraction, offers broad practical applications. In clinical decision support, it speeds up case reviews by extracting relationships between body sites and symptoms and links case conditions to historical treatments for treatment plan matching. For quality and standardization, tracking treatment-indication relationships ensures consistent use of medical terminology across doctors. In patient care, it monitors the relationship between interventions and symptom changes, providing structured information to improve communication between physicians. In education and training, it helps doctors recognize common patterns in coronary angiography results.

6. Conclusion

This study presents a contrastive learning-based joint entity and relation extraction method, which relies on data augmentation-driven contrastive learning and a cascade decoding mechanism. This strategy not only enhances the model's ability to extract triple relationships under limited data conditions, thereby mitigating the performance degradation caused by data scarcity but also significantly improves the handling of overlapping relationships. The model employs a CKBERT-based text encoder, which incorporates linguistic and external knowledge to enhance semantic understanding. Additionally, the feature enhancement module BMG, based on an attention mechanism, is introduced to bolster semantic feature information. We propose a multi-strategy integrated data augmentation method that improves the model's relationship extraction performance in low-sample scenarios. By mapping positive and negative sample pairs into a unified contrastive learning space for contrastive loss computation, the model's comprehension of key information is enhanced, thereby increasing its ability to distinguish relationships. This provides robust support for differentiating complex relationships. Experimental results on our constructed CACMeD dataset and the publicly available DuIE dataset demonstrate the effectiveness of our model. In the future, we would like to further investigate the problem of modeling relational extraction in rare and complex relationships.

CRediT authorship contribution statement

Yetao Tong: Writing – original draft, Validation, Software, Methodology. **Jijun Tong:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Shudong Xia:** Resources, Data curation. **Qingli Zhou:** Supervision, Resources, Funding acquisition, Data curation. **Yuqiang Shen:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Key R&D Program of Zhejiang (2024C03055) and the Key R&D Program of Zhejiang (2024C03270).

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